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DESCRIBE THE TYPES OF COMPUTER SYSTEMS AND ASSOCIATED HARDWARE CONFIGURATIONS

US 14921

SECTION B: SUMMATIVE ASSESSMENT

TASK 1: REPORT ON PAST, PRESENT & FUTURE COMPUTER HARDWARE CONFIGURATIONS

1. INTRODUCTION

Computer hardware has evolved to meet changing business and personal needs. This report describes three past, three present, and three future hardware configurations. Each configuration is discussed in terms of characteristics, performance, and environmental requirements.

2. PAST HARDWARE CONFIGURATIONS

A. Mainframe - IBM System/360 [1960s]

Characteristics: Room-sized, supported 100s of users, batch processing, punched 7cards, limited RAM.

Justification: Categorised as mainframe due to centralized processing for large organisations.

Performance: Low speed by today's standards but high throughput for payroll and banking.

Environmental Requirements: Dedicated computer room, raised floor, industrial cooling, backup generator.

B. Minicomputer - DEC PDP-11 [1970s]

Characteristics: Medium size, supported 10-50 users, used in departments.

Justification: Mid-range because it was between mainframes and PCs in price and power.

Performance: Good for scientific and business tasks. Less scalable than mainframes.

Environmental Requirements: Standard office with air conditioning.

C. Microcomputer - IBM PC [1981]

Characteristics: Desktop, single-user, 640KB RAM, floppy disk, DOS.

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Justification: Microcomputer for individual use at home and office.

Performance: Basic tasks like word processing and spreadsheets.

Environmental Requirements: Normal home/office environment, no special requirements.

3. PRESENT HARDWARE CONFIGURATIONS

A. Personal Computer / Laptop [2024-2026]

Characteristics: Multi-core CPU, 16-32GB RAM, 1TB SSD, Wi-Fi, single-user.

Justification: Microcomputer for personal productivity and business.

Performance: Fast multitasking, supports Office, design software, and light gaming.

Environmental Requirements: Normal office, surge protector, room temperature.

B. Business Server

Characteristics: Rack-mounted, 64GB+ RAM, RAID storage, redundant power.

Justification: Mid-range system used to serve company networks.

Performance: High reliability, supports 200+ users, 24/7 uptime.

Environmental Requirements: Data center, cooling, UPS, fire suppression.

C. Smartphone

Characteristics: Pocket size, 8-core CPU, 12GB RAM, 256GB storage.

Justification: Microcomputer for mobile personal computing.

Performance: Optimized for apps, battery life, and portability.

Environmental Requirements: No special environment. Sensitive to water and heat.

4. FUTURE HARDWARE CONFIGURATIONS

A. Quantum Computer

Characteristics: Uses qubits, operates at near absolute zero.

Justification: Future technology for research, AI, and cryptography.

Performance: Solves complex problems in minutes instead of years.

Environmental Requirements: Cryogenic cooling, vibration-free lab.

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B. Neuromorphic Computer

Characteristics: Mimics human brain, uses spiking neural networks.

Justification: Future AI hardware for learning and pattern recognition.

Performance: Very low power, excellent for real-time AI.

Environmental Requirements: Standard environment, energy efficient.

C. Edge AI Device

Characteristics: Small chip that runs AI locally in devices.

Justification: Future trend to reduce cloud dependency for IoT and cars.

Performance: Fast local decision making with low latency.

Environmental Requirements: Can operate in factories, vehicles, outdoors.

5. CONCLUSION

Hardware has moved from large centralized mainframes to personal devices. The future points toward AI, edge, and quantum computing with lower power and higher specialization.

TASK 2: SYSTEM ADMINISTRATOR INTERVIEW REPORT - PILSON ENGINEERING

1. COMPANY BRIEF HISTORY

Company Name: Pilson Engineering

Location: Johannesburg, Gauteng, South Africa

Established: 2010

Services: Mechanical engineering, fabrication, CNC machining, industrial maintenance, and project management for mining and manufacturing clients.

Number of Employees: ±45 staff across workshop, office, and site teams

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2. INTERVIEWEE

Name: Mr. J. Dlamini

Position: System Administrator / IT Support

Date of Interview: 19 August 2026

Q1: What software applications does the company use?

A: We use AutoCAD and SolidWorks for engineering design, Sage 300 for accounting and payroll, Microsoft 365 for email and documents, Pastel Partner for inventory, and Kaspersky for antivirus. We also use a workshop job-card system called Workshop Butler.

Q2: Why were these specific applications selected?

A: AutoCAD and SolidWorks are industry standards for engineering, so clients expect drawings in those formats. Sage 300 was chosen because it handles payroll for 40+ staff and integrates with SARS. Microsoft 365 was selected for cloud backup and remote access. Kaspersky for security because we deal with client drawings.

Q3: Which category does each application fall under?

A:

- System Software*: Windows 11 Pro, Windows Server

- Business Software: Sage 300, Pastel Partner

- Application Software: Microsoft 365, Workshop Butler

- Engineering/Scientific Software*: AutoCAD, SolidWorks

- Security Software: Kaspersky Endpoint

Q4: What are the hardware and environmental requirements?

A: Engineering PCs need i7/i9 processors, 32GB RAM, and dedicated Nvidia graphics for CAD. Servers need 64GB RAM, RAID storage, and UPS backup. Workshop PCs need to be dust-proof. Office needs aircon and stable internet. We also have a generator for load shedding.

Q5: How do you handle software updates and security?

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A: Windows and Microsoft 365 update automatically every month. Sage and AutoCAD we update quarterly after testing. Kaspersky scans daily. All important files are backed up to OneDrive and an external NAS weekly.

Q6: What challenges have you faced with the current setup?

A: Load shedding is the biggest problem. Also, CAD files are very large so internet speed for sharing with clients is slow. Licensing costs for SolidWorks are expensive.

Q7: How many users does the system support?

A: About 35 office users and 10 workshop terminals. Server handles file sharing and printing for all of them.

Q8: Are there any plans to upgrade software or hardware soon?*

A: Yes. In 2027 we plan to move Sage to the cloud version and upgrade all design PCs to RTX graphics cards. We're also looking at adding project management software.

Q9: What advice would you give to IT students?

A: Learn networking, backups, and how to support engineering software. In engineering companies IT must understand both office and workshop needs. Also learn about cybersecurity because client designs are confidential.

Q10: How do you decide to purchase new software?*

A: We check 3 things: 1. Does it integrate with what we already use? 2. Is there local South African support? 3. What is the total cost including training. We never buy just because it's new.

4. SUMMARY OF FINDINGS

Pilson Engineering uses a mix of system, business, and engineering application software. The choice of software is driven by industry standards, cost, and support. Environmental needs are higher than a normal office because of dust in the workshop and power issues. This shows how different business types need different categories of software.